

PROGRAM-5

TITLE: Demonstrate CI Integration by running Jenkins in a Docker Container

Project Structure

program-5

Dockerfile

docker-compose.yml

Step 1: create a Dockerfile

```
GNU nano 6.2 Dockerfile
FROM jenkins/jenkins:lts
USER root
RUN apt-get update && apt-get install -y docker.io
USER jenkins
```

Step 2: create a docker-compose.yml

```
GNU nano 6.2 docker-compose.y
version: '3.8'
services:
jenkins:
build: .
container_name: jenkins-ci
ports:
- "8080:8080"
- "50000:50000"
volumes:
- jenkins_home: /var/jenkins_home
- /var/run/docker.sock: /var/run/docker.sock
volumes:
jenkins_home
```

Step 3: Build a docker image

> docker build -t program5 .

```
1RV24MC018_Ananya_h@user-ThinkPad-E480:~/labprograms/Program-5$ docker build -t program5 .
[+] Building 4.5s (6/6) FINISHED
=> [internal] load build definition from Dockerfile
=> => transferring dockerfile: 136B
=> [internal] load metadata for docker.io/jenkins/jenkins:lts
=> [internal] load .dockerignore
=> => transferring context: 2B
=> [1/2] FROM docker.io/jenkins/jenkins:lts@sha256:7b1c378278279c8688efd6168c25a1c2723a6bd6f0420beb5ccefabee3cc3bb1
=> CACHED [2/2] RUN apt-get update && apt-get install -y docker.io
=> exporting to image
=> => exporting layers
=> => writing image sha256:b803d11efefe94031a72c02c0f0d463217252f77ba04a31b3864d77dd4c29612
=> => naming to docker.io/library/program5
```

Step 4: Run a docker image

> docker run -d -p 8080:8080 -p 50000:50000 program5

```
1RV24MC018_Ananya_h@user-ThinkPad-E480:~/Labprograms/Program-5$ docker run -d -p 8080:8080 -p 50000:50000 program5
91cd69517aedf1e67f4e6c1df96bffe8305f6aa9b82b3b3b6ae5c3d03962410d
1RV24MC018_Ananya_h@user-ThinkPad-E480:~/Labprograms/Program-5$ docker ps -a
dockerps: command not found
1RV24MC018_Ananya_h@user-ThinkPad-E480:~/Labprograms/Program-5$ docker ps -a
CONTAINER ID   IMAGE          COMMAND                  CREATED        STATUS        PORTS
91cd69517aed   program5       "/usr/bin/tini -- /u...  9 seconds ago Up 9 seconds  0.0.0.0:8080->8080/tcp, [::]:8080->8080/tcp, 0.0.0.0:50000->50000/tcp, [::]:50000->50000/tcp
0c36a0008fc8   program4       "docker-entrypoint.s...  6 minutes ago Up 6 minutes  0.0.0.0:3000->3000/tcp, [::]:3000->3000/tcp
```

Step 5: open the browser and check <http://localhost:8080>

